

Salient Characteristics for a Mobile Fire Pump Testing & Training Unit

1. General Description

The unit shall be a **mobile fire pump testing and training system** capable of performing true pump tests from draft, creating a suction lift requiring the use of a primer in accordance with **NFPA 1911 (2017 Edition)**. The system must support pump testing up to **3,000 GPM** and allow training operations using recycled water.

The document states the system “*performs a true pump test from draft creating a suction lift requiring the use of a primer as per NFPA 1911-2017 Edition*” and is designed for “*flowing clean clear water... up to 3000 GPM.*”

2. Transporter Requirements

The mobile transporter shall meet the following minimum characteristics:

2.1 Construction & Dimensions

- Purpose-built, heavy-duty aluminum transporter engineered for low profile, proper weight distribution, and safe towing.
- Minimum **80-inch wide** heavy-duty main frame with **½-inch extruded bed railing**.
- Overall dimensions (empty):
 - Weight: ≤ **6,500 lbs**
 - Height: ≤ **87 inches**
 - Width: ≤ **102 inches**
 - Length: ≤ **27 feet**

2.2 Axles & Support

- Two **7,000-lb rubber torsion axles** with electric brakes; mounted to a steel tube subframe.
- Four **7,000-lb screw-type stabilizing jacks** (two per side).
- One **12,000-lb tongue jack**.

2.3 Storage & Access

- Two front aluminum storage compartments, each approx. **30" W × 29" D × 44" H**, with three-point latching doors.

- Minimum **24-inch** space between compartments and reservoir front for spare tire and cooling access.

2.4 Lighting & Safety

- LED stop, tail, turn, ICC, and work lights.
- Rear crossbar with elevated LED lighting.
- DOT-approved reflective tape on all sides.

2.5 Flooring & Fenders

- **2-inch extruded aluminum tongue-and-groove flooring** (wood or steel not acceptable).
- Aluminum tread-plate fenders with integrated steps.

2.6 Hitch

- Adjustable **2-5/16" ball** or **pintle-type** towing connection.

3. Water Reservoir Requirements

A minimum **3,000-gallon capacity** reservoir meeting the following:

3.1 Construction

- Heavy-duty poly material for all internal and external surfaces.
- “T-shaped” design, securely mounted to prevent shifting while allowing flex with transporter.

3.2 Drafting Pit & Internal Flow Control

- Integrated drafting pit with **four anti-swirl plates**.
- Internal cavitation-control baffles made of **½-inch or thicker poly**, welded to withstand high-volume flow.
- System designed to eliminate air disturbance and maintain proper draft depth.

The document notes the reservoir includes a section that “*removes cavitations/disturbance and air from the water... preventing water disturbance and air swirls causing the apparatus to lose prime.*”

3.3 Cooling & Ventilation

- Integrated water-cooling system with:

- Two **2½-inch inlets**
- One **4-inch warm-water discharge** with gate valve
- Front and rear ventilators with hinged polypropylene doors and hold-open devices.

3.4 Additional Features

- UV-rated net cover for open reservoir area.
- Removable inspection plates at all four corners.
- Rub rails on all sides.
- Rear sump drain with **2½-inch gate valve**.
- Two liquid sight gauges at rear drafting pit.

4. Drafting System

4.1 Drafting Tubes & Swivels

- Two **6-inch drafting tubes** with heavy-duty **45° aluminum 360° swivels** (PVC not acceptable).
- Each swivel equipped with grease fittings and **¾-inch vacuum breaker** controlled at rear.

4.2 Hard Suction Hose

- Two **17-ft sections of 6-inch clear hard suction hose**, preconnected.
- Two **10-ft sections of 6-inch** clear hard suction hose.
- Two **10-ft sections of 2½-inch** clear hard suction hose.
- Aluminum hose trays with hold-down straps.
- Full-length aluminum support rails around reservoir perimeter.
- Fold-out support bars (left and right) for weight distribution during testing.

5. Flow Measurement & Discharge System

5.1 Inlet Manifolds

- Two **4-inch aluminum manifolds**, one per side, color-coded.
- Each equipped with:
 - **5-inch Storz male connection**

- Three **45° 2½-inch swivels**
- Quarter-turn drain valves

5.2 Pitot Compartments

- Two large pitot compartments with hinged poly doors, splash guards, diffuser tips, and quick-disconnect fittings.
- Chrome grab handles.
- Color-coded left/right.

5.3 Pressure Relief

- Two NFPA-compliant pressure relief valves (drip at 120 PSI, full discharge at 135 PSI).

6. Monitoring & Test Instrumentation

6.1 Master Monitor Station

- Portable monitoring station approx. **20" W × 18" H × 6" D**.
- Aluminum tripod mount.
- Gauges (all glycerin-filled, freeze/clog resistant, calibrated with certificates):
 - Two **0–160 PSI pitot gauges**
 - One **0–30" vacuum gauge**
 - One **0–400 PSI pressure gauge**
- Color-coded bezels and hose fittings.
- Two **30-ft high-pressure pitot hoses**.
- Two **10-ft high-pressure hoses** for UL test ports.
- Integrated flow and conversion charts.

7. Hose Storage & Miscellaneous Equipment

- Rear storage compartment approx. **35½" W × 33¼" H × 28" D**, ventilated, with three-point latch.
- Capacity for six 50-ft sections of 3-inch hose plus tools.

8. Integrated Fire Hose Tester

- Gas-powered hose tester with minimum **5.5-HP engine**.
- Capable of testing hoses up to **6-inch diameter**.
- Stainless steel manifold with four independently controlled ¼-turn valves.
- Master gauge (0–600 PSI) and four individual 0–600 PSI gauges.
- Includes NST caps with bleeder valves and Storz adapters.
- Permanently mounted and preconnected to reservoir.

9. Included Accessories

The system shall include, at minimum:

- Mounted spare tire with UV-rated cover
- Full-size UV-rated protective cover for entire unit
- Gauge test kit (pitot, pressure, vacuum gauges with fittings)
- High-pressure pump test kit for high-pressure apparatus
- Six 50-ft sections of 3-inch double-jacket hose (600-PSI test)
- Multiple NST adapters (6"→5", 6"→4½", 6"→4", 5" Storz→2½")
- 30° 6-inch waterway elbow for strain relief
- Rubber mallet, T-handle wrench, spanner wrenches, hydrant wrench

10. Other Equipment

- Handline nozzle tester (30–500 GPM range)
- Gauge calibration services